

An autochthonous geological model for the eastern Andes of Ecuador[☆]

Warren T. Pratt^{a,*}, Pablo Duque^b, Miguel Ponce^c

^a*Specialized Geological Mapping Ltd., Garmouth, Moray IV32 7NT, UK*

^b*Departamento de Geología, Escuela Politécnica Nacional, Casilla 17-01-2759, Quito, Ecuador*

^c*International Minerals Corporation, Calle Jimenez de la Espada 195 y Av. G. Suarez, Quito, Ecuador*

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Abstract

We describe a traverse across the Cordillera Real and sub-Andean Zone of Ecuador, poorly known areas with very little detailed mapping and very little age control. The spine of the Cordillera comprises deeply eroded Triassic and Jurassic plutons, the roots of a major arc, emplaced into probable Palaeozoic pelites and metamorphosed volcanic rocks. The W flank comprises a Jurassic (?) submarine basaltic–andesitic volcanic sequence, which grades up into mixed Jurassic/Cretaceous volcanic and sedimentary rocks of the Inter-Andean Valley. The sub-Andean Zone, on the E flank of the Cordillera, comprises a newly recognized Cretaceous basin of cleaved mudrocks, quartz arenites and limestones. East of the syndepositional Cosanga Fault, the Cretaceous basin thins into a condensed sequence that is indistinguishable from the rocks of the adjacent hydrocarbon-bearing Oriente Basin. The principal penetrative deformation of the Cordillera Real was probably latest Cretaceous/Palaeocene. It telescoped the magmatic belts, but shortening was largely partitioned into the pelites between plutons. The plutons suffered inhomogeneous deformation; some portions completely escaped tectonism. The pelites conserve two foliations. The earliest comprises slaty cleavage formed under low- or sub-greenschist conditions. The later is a strong schistosity defined by new mica growth. It largely transposed and obliterated the first. Both foliations may have developed during a single progressive deformation. We find inappropriate recent terrane models for the Cordillera Real and sub-Andean Zone of Ecuador. Instead we find remarkable similarities from one side of the Cordillera to the other, including a common structural history. In place of sutures, we find mostly intrusive contacts between major plutons and pelites. Triassic to Cretaceous events occurred on the autochthonous western edge of the Archaean Guyana Shield. The latest Cretaceous–Paleocene deformation is interpreted as the progressive collision of an oceanic terrane(s) with the South American continent. Young fault movements have subsequently juxtaposed different structural levels through the Cordillera Real orogen.

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Keywords: Northern Andes; Cordillera Real; Ecuador; Subduction; Strain partitioning; Blue quartz; Hydrocarbons

[☆] Permission should be sought before entering the Río Pastaza beneath the Agoyán dam. Periodic opening of the floodgates causes a sudden and dangerous rise in water level.

* Corresponding author.

E-mail addresses: info@geologicalmapping.com (W.T. Pratt), pduque@mac.com (P. Duque), geomigue@yahoo.com (M. Ponce).

1. Introduction

The Andes are the classic example of a mountain range built above a subduction zone. Patterns of seismicity and volcanism leave little doubt that the Pacific Plate is consumed beneath South America and that this has happened since at least the Late Jurassic (Gansser, 1973; Lonsdale, 1978). The Andes have probably grown by compression, uplift, intrusion, crustal thickening, and volcanism. However, a large part of the northern Andes, mainly oceanic crust, is regarded as allochthonous, accreted during the Cretaceous and Tertiary (Malfait and Dinkleman, 1972; Feininger and Bristow, 1980; McCourt et al., 1984; Feininger, 1987; Jaillard et al., 1990; Reynaud et al., 1999) (Fig. 1). Strike-slip is frequently invoked in these accretions (Aspden and Litherland, 1992; Kerr et al., 2002). In the Central and Eastern Cordilleras of

Colombia, allochthonous terranes have been suggested in Palaeozoic (Forero Suarez, 1990; Restrepo-Pace, 1992), even Precambrian strata (Irving, 1975). The Huancabamba Deflection of northern Perú, where the Andes swing from NNW to NNE (Gerth, 1955; Ham and Herrera, 1963), probably marks the limit of the allochthons (Mourier et al., 1988); exotic Jurassic/Cretaceous terranes are not recognized farther S.

The principal topographic components of Ecuador are shown in Fig. 1. They comprise, from W to E: (1) a coastal plain, underlain by Cretaceous oceanic basalts (Piñón Formation; Tschopp, 1948; Goossens and Rose, 1973); (2) the Cordillera Occidental, which comprises Cretaceous pillowed basalts, historically correlated with the Piñón (Sauer, 1965; Baldock, 1982), and an Eocene island arc (Henderson, 1979; Eguez, 1986); (3) the Inter-Andean Valley, flanked by active and Recent

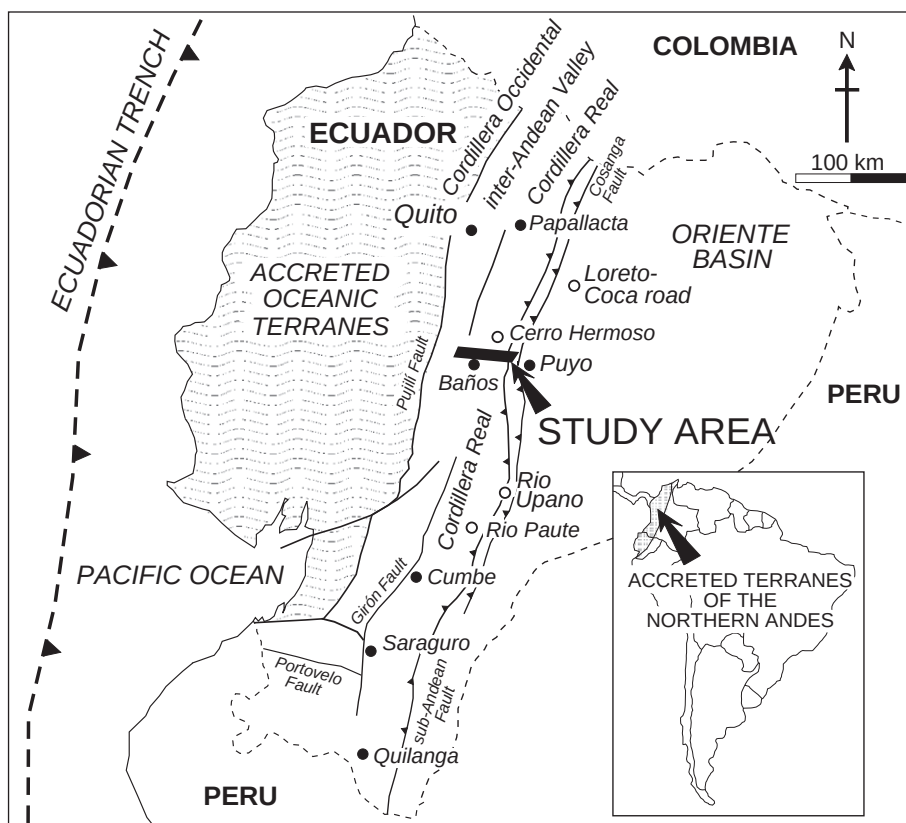


Fig. 1. Simplified map of the principal structural components of Ecuador.

volcanoes, such as Cotopaxi; (4) the Cordillera Real, which comprises N-striking Precambrian to Mesozoic metamorphic rocks; (5) the sub-Andean Zone, a belt of cleaved and faulted Jurassic/Cretaceous formations; (6) the Oriente Basin, which comprises flat-lying Cretaceous hydrocarbon-bearing formations concealed by Neogene to Recent deposits derived from the Andes; (7) the Archaean Guyana Shield. This simplistic picture is complicated by the recent recognition of additional oceanic and island arc terranes in the coastal plain and Cordillera Occidental (e.g. Reynaud et al., 1999; Hughes and Pilatasig, 2002).

In Ecuador, the boundary between accreted terranes and S American continental crust is not clear. The coastal plain coincides with one of the world's largest onshore positive Bouguer gravity anomalies (Feininger and Seguin, 1983). It is thought to be a fragment of exotic, or allochthonous, oceanic plateau crust (e.g. Goossens and Rose, 1973; Reynaud et al., 1999). Oceanic crust in the Cordillera Occidental is thought to be allochthonous (e.g. Feininger and Bristow, 1980), like that of Colombia (McCourt et al., 1984; Bourgois et al., 1987). Most estimates place the limit of accreted oceanic crust on the W flank of the Inter-Andean Valley (e.g. Hughes and Pilatasig, 2002). However, it has been proposed that the Cordillera Real is also allochthonous with respect to the Guyana Shield. The Cordillera has been modeled as a Californian-type collage of narrow Mesozoic terranes, introduced by dextral strike-slip or dextral transpression (Aspden and Litherland, 1992; Litherland et al., 1994). This terrane template features in most subsequent work in the Cordillera Real and in the World Bank metal exploration manuals for Ecuador (PRODEMINCA, 2000).

We present here the first test of the terrane model. It is based on detailed structural mapping of the best exposed transect, during various field sessions between 2001 and 2003, and is backed up by work throughout the Cordillera Real and sub-Andean zone during 5 years of precious metal and hydrocarbon exploration.

2. Previous research

Early studies of the Cordillera Real include the geological map and memoir of Sauer (1957, 1965), and petrographical and structural transects by Herbert

(1977, 1983) and Trouw (1976). Baldock (1982) compiled Institute of Geological Sciences work and oil company data into a national geological map (DGGM-IGS, 1982). Feininger (1982) did a preliminary compilation of the metamorphic geology of Ecuador. The British Geological Survey (BGS) mapped the Cordillera Real between 1986 and 1990 (Aspden and Litherland, 1992; Litherland et al., 1994). The BGS work, supported by isotopic dating, profoundly changed the map representation of the metamorphic rocks of Ecuador. Most contacts between units were depicted as faults and the interpretation was radical, identifying a series of suspect terranes, including island arc, passive margin and ensialic basins. From W to E the terranes are: Guamote (continental), Alao-Paute (island arc), Loja (continental), Salado (island arc) and Amazonic (continental) (Fig. 2). The respective sutures are the Peltetec, Baños, Llanganates and Cosanga faults.

3. Study area and general geology

We mapped approximately 35 km along the Río Pastaza, centred on the town of Baños. This is the most accessible corridor across the Cordillera Real and lies midway between Colombia and Perú. Despite the short distance, it crosses five supposed terranes and four sutures (Fig. 2). Several 1:50,000 scale geological maps cover the section (Kennerley, 1971; DGGM, 1980). Metamorphosed Triassic and Jurassic plutons dominate, separated by screens of metamorphosed sedimentary and volcanic rocks. The most important structure is the sub-Andean Fault, a major reverse fault that brings metamorphic rocks over the sub-Andean Zone, a thrust belt of cleaved Cretaceous mudstones and sandstones. Apart from fossiliferous formations in the sub-Andean Zone and Inter-Andean Valley (Fig. 2), and intrusions dated by BGS, there is little age control in the Cordillera Real. The metamorphic rocks are locally concealed by Recent lavas from Tungurahua, an active volcano near Baños (Fig. 2).

4. Cordillera Real stratigraphy

Litherland et al. (1994) recognized the difficulty of establishing a coherent lithostratigraphy in an area

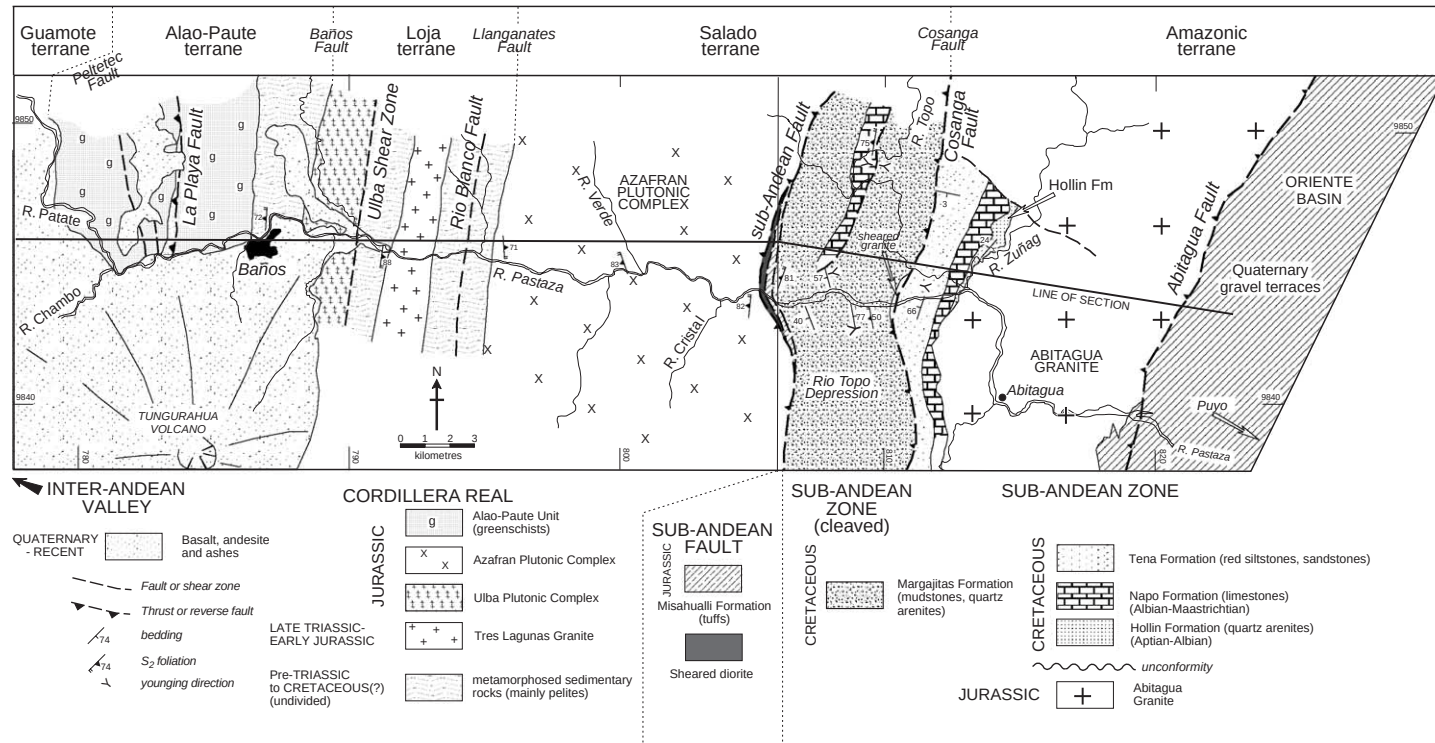


Fig. 2. Geological map of the Baños-Puyo corridor. The terranes and terrane boundaries proposed by Litherland et al. (1994) are shown in the top bar. Minor amphibolites, granites and Recent lavas are omitted. See Figs. 4 and 5 for details.

with such poor age control. They divided the Cordillera Real into ‘lithotectonic divisions’ separated by major faults. This required grouping dissimilar lithologies into single units. For example, the Alao-Paute unit (Alao-Paute terrane) (Fig. 2) was defined principally as meta-basaltic and meta-andesitic greenstones, with meta-sediments and marbles. Likewise, the Upano unit (Salado terrane) was defined principally as andesitic greenstone, with subordinate meta-sedimentary rocks. We find this classification difficult to sustain in the Baños corridor because of evidence of stratigraphic links between the divisions. Furthermore, the lithotectonic divisions are locally represented entirely by the subordinate lithology. For example, the Upano unit in the Baños corridor contains no meta-andesites, only mudstones. Indeed, instead of being lithological, the main basis for the divisions is geographic, that is, their location between major faults. Since some of these faults are in fact intrusive contacts, the basis of the classification is further undermined.

Considering the relatively small volume of meta-sedimentary rocks in the Baños corridor, and the uncertainties about their correlation within, and outside, the Cordillera Real, we have simply combined all outcrops between the Inter-Andean Valley and the sub-Andean Fault as ‘metamorphosed sedimentary rocks’ (Figs. 2 and 3).

4.1. Meta-sedimentary rocks (pre-Late Triassic to Cretaceous?)

Baldock (1982) grouped the meta-sedimentary strata of the Cordillera Real into two geographical units: the Zamora Series in the S, and Llanganates Series in the N (DGGM-IGS, 1982). We believe there are at least two deformed and metamorphosed mudrock-dominated sequences of different ages within the Cordillera Real. However, we cannot identify precisely the contact between them because of metamorphism and strong polyphase deformation. An older (Palaeozoic) sequence is implied by the intrusive contact of the Tres Lagunas Granite (Late Triassic) with pelites. The younger is suggested by Jurassic/Cretaceous microfossil faunas (Litherland et al., 1994).

The older pelites occupy the core of the Baños corridor and are the highest-grade metamorphic rocks,

reaching high greenschist, possibly low amphibolite, facies. They comprise dark grey schists with quartz+muscovite+albite±biotite±garnet±graphite±epidote. Bands of coarse granoblastic quartz between 1 and 20 mm thick are common between graphitic and micaeous domains. Close to the Tres Lagunas Granite and Ulba Plutonic Complex the rocks are tougher, coarser-grained schists with common albite and garnet porphyroblasts. A 30-m-thick marble, with traces of serpentine minerals, is exposed within the older pelites at the mouth of the Agoyán hydroelectric tunnel [793660 9845150].

The younger pelites are mainly black to dark grey, non-calcareous to strongly calcareous phyllites with a wide variety of meta-volcanic rocks and metamorphosed clastic sedimentary rocks with volcanic/igneous material. Disseminated pyrite is ubiquitous and gives the exposures a rusty appearance, e.g. the bridge at Sauce (Fig. 4). Sigmoidal and folded quartz veins are widespread (Figs. 4A,B and 10A,C). Over 200 m (stratigraphic thickness) of meta-sandstones, meta-conglomerates, and graphitic phyllites, with abundant chloritoid, crop out at the confluence of the ríos Blanco and Pastaza (Fig. 5). Beds of poorly sorted conglomerates, up to 1.5 m thick, contain clasts up to 150 mm long (Fig. 6). Clasts are mainly metamorphosed fine-grained andesitic(?) igneous rock, with scattered vein quartz, cream granitoid and distinctive grains of blue quartz. The massively bedded meta-sandstones have a streaky, disturbed appearance (bioturbation?) and contain common mudstone rip-up clasts.

Graphitic phyllites exposed immediately N of Baños (Fig. 4) include beds of variegated light green to grey schist, probably former tuffs or tuffites. The colours reflect the content of graphite versus mica/chlorite. Minor calc-schists, metaquartzites, biotite+garnet schists and greenschists also occur. Rare sedimentary grading and sole marks indicate westward younging and suggest that these rocks mark an upward transition into the massive Alao-Paute Unit greenschists to the W.

The hanging wall of the La Playa Fault (Fig. 4) comprises several hundred metres of graphitic calcareous phyllite with disseminated pyrite. Light green calc-schists (calcite+talc+chlorite±fuchsite) are common; scattered marbles up to 5 m thick also occur, e.g. [783440 9845230]. These strata lie along strike from,

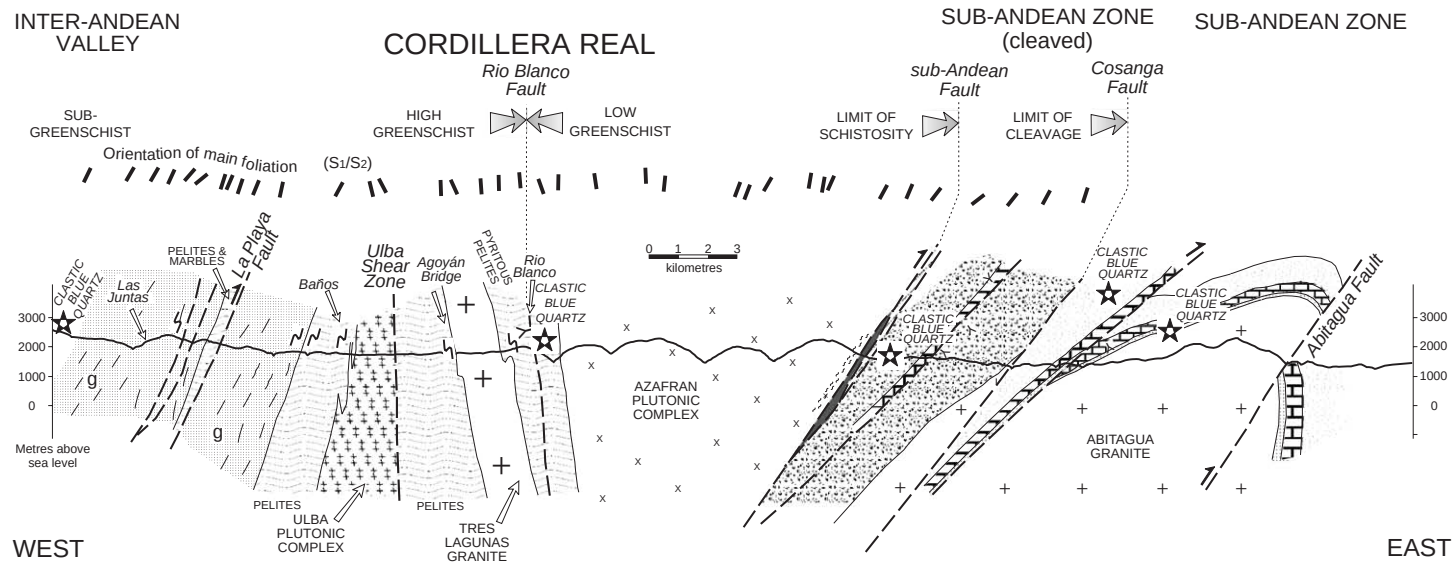
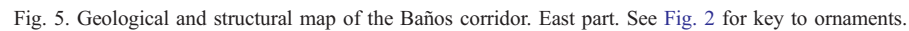


Fig. 3. Geological cross-section along the line indicated in Fig. 2. Quaternary and Recent volcanic rocks, minor granites and amphibolites are omitted. No vertical exaggeration. See Fig. 2 for key to ornaments.



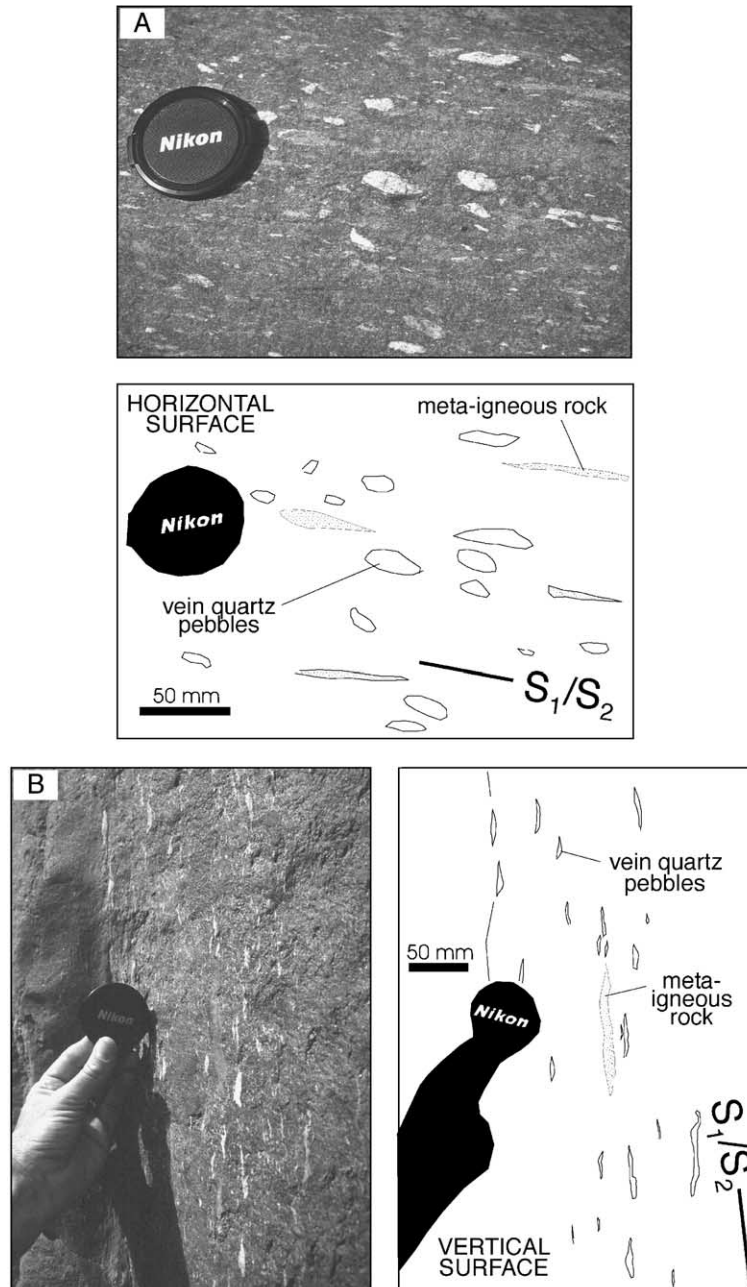


Fig. 6. Meta-conglomerate bed. Rio Blanco [794700 9845225]. The long axes of the pebbles plunge 70°N.

and closely resemble, metamorphosed calcareous rocks at Cerro Hermoso, to the N of Baños (Fig. 1). Many authors correlate Cerro Hermoso with the non-metamorphosed Napo Formation (Albian–Maastrichtian) of the Oriente Basin (Wolf, 1892; Tschopp,

1956; Bristow and Hoffstetter, 1977). Litherland (1988) noted their bituminous character. However, since the Baños exposures probably lie stratigraphically beneath the Alao-Paute Unit, a Jurassic age seems more likely.

4.2. Alao-Paute Unit (Jurassic?) (Litherland et al., 1994)

This is the greenschist component of the ‘Alao-Paute unit’ of Litherland et al. (1994). It crops out in the W, as two steeply inclined, possibly fault-repeated, bodies either side of the La Playa Fault (Figs. 3 and 4). There are few sedimentary, volcanic, or structural younging indicators, but it appears to young towards the W. The dominant lithology is a moderately fissile greenschist. Massive meta-basalts at Las Juntas (Fig. 4) preserve original porphyritic texture, with plagioclase phenocrysts. The locality also shows meta-gabbro dykes within meta-basalt, implying a syngenetic relationship. Pillowed basalts [781400 9844600] and pillows with interpillow graphitic phyllite occur nearby [784290 9845380]. The pillows have internal cooling cracks and porphyritic texture.

We interpret the Alao-Paute Unit as a submarine volcanic sequence. Limited geochemical samples from the equivalent (?) San Francisco meta-volcanics, near Cuenca, in southern Ecuador (Bristow, 1973), imply an oceanic island arc origin (Litherland et al., 1994), but the rocks are altered and the geochemical data therefore equivocal. The stratigraphic correlation is also uncertain. K–Ar dates from the unit are Early Cretaceous, but fossils from adjacent sedimentary rocks suggest a Late Jurassic age (Litherland et al., 1994).

4.3. Maguazo and Cebadas units (Jurassic?/Cretaceous) (Litherland et al., 1994)

The Maguazo unit comprises metamorphosed volcanic and sedimentary rocks and lies W of, and stratigraphically above, the Alao-Paute unit. Litherland et al. (1994) mapped both contacts as faults, including, in the W, a terrane boundary (the Peltetec fault or Peltetec ophiolite belt). However, we find a complete stratigraphic transition from Alao-Paute Unit greenschists in the E, to sedimentary rocks (Cebadas unit) in the Inter-Andean Valley (Figs. 2 and 4). The transition coincides with a pronounced decline in metamorphic grade and foliation intensity. Minor faults occur, but there is no significant tectonic break. West of Las Juntas, there is a clear transition from massive Alao-Paute Unit greenschists to interbedded meta-volcanic and meta-sedimentary rocks (Maguazo

unit of Litherland et al., 1994). Farther W, at Pungapi [781000 9846900] (Fig. 4), meta-basalts are intercalated with thinly bedded black phyllites and meta-greywackes. Nearby, weakly metamorphosed andesitic crystal and lapilli tuffs crop out in the Río Patate [779300 9848800]. A road cut W of the river shows weakly metamorphosed andesitic tuffs interbedded with black graphitic and calcareous phyllites, quartzites and meta-greywackes up to 0.75 m (Fig. 4). The latter contain detrital blue quartz, a typical feature of sedimentary rocks in the Inter-Andean Valley. Nearby exposures of black slates, at Patate, yielded Jurassic?/Cretaceous spores (Cebadas unit, Litherland et al., 1994). The slates are interbedded with quartzites with blue quartz grains.

The stratigraphic transition from the Alao-Paute greenschists to the Inter-Andean Valley sedimentary rocks is important: it supports a Jurassic age for the Alao-Paute Unit and effectively rules out a terrane boundary in this position. If such a boundary exists, it must lie farther W, beneath the Inter-Andean Valley.

5. Oriente Basin and sub-Andean Zone stratigraphy

5.1. Misahualli Formation (Jurassic) (original definition by Wasson and Sinclair, 1927)

Patchily exposed in the sub-Andean Zone, this is part of a belt of Jurassic calc-alkaline volcanism that stretches from northern Chile to Colombia (Romeuf et al., 1995). Isotopic dates include 132 Ma (K–Ar; Hall and Calle, 1982), 162 ± 2 Ma ($^{40}\text{Ar}/^{39}\text{Ar}$; Spikings et al., 2001) and 172.3 ± 2.1 Ma ($^{40}\text{Ar}/^{39}\text{Ar}$; Romeuf et al., 1995). We include the sporadic exposures of volcanic rocks in the sub-Andean Zone of the Baños corridor with this formation. This is because adjacent exposures comprise Cretaceous formations (see Section 5.2) and large outcrops of Misahualli Formation occur a small distance to the N (Litherland et al., 1994). In the Baños corridor, a wedge of andesitic/dacitic welded ash-flow tuff, 100 m wide, crops out in the footwall of the sub-Andean Fault [805600 9844090]. River float suggests that roof pendants also occur over the Abitagua Granite (Fig. 2).

5.2. *Hollín, Napo and Tena formations (total ca. 1250 m) (Aptian to Maastrichtian/Palaeocene) (Wasson and Sinclair, 1927; Ribadeneira, 1942)*

Jaillard et al. (1997) provide the most recent stratigraphic classification of the Cretaceous rocks of the Oriente Basin. Within the Baños corridor, non-metamorphosed, bituminous strata, indistinguishable from the Oriente, crop out along the W flank of the Abitagua Granite (Figs. 2 and 3).

The Hollín Formation (Albian–Aptian) represents a regionally important Early Cretaceous marine transgression in the northern Andes (e.g. Benavides, 1956; Jaillard et al., 2000). It is the main hydrocarbon reservoir of the Oriente, Putumayo (Colombia) and Marañón (Perú) basins. Within the sub-Andean Zone of Ecuador it unconformably overlies Jurassic granitoid plutons or the Misahuallí Formation. In oil-soaked exposures on the Loreto-Coca road [194720 9922202] (Fig. 1), NE of the Baños corridor, it truncates steeply inclined Misahuallí welded tuffs. The formation was derived from the Guyana Shield (Jaillard et al., 2000), supported by foreset-derived palaeocurrent data from Loreto-Coca, which are toward the W.

The Hollín Formation in the Baños corridor comprises about 60 m of well-sorted quartz arenites, siltstones, bioturbated carbonaceous mudstones, and thin coals (up to 0.2 m). The arenites are buff, where oxidized, or dark grey where they conserve disseminated pyrite. The quartz arenites contain about 1% bright blue quartz grains (also in the Loreto-Coca exposures). The basal few metres comprise conglomerates rich in vein quartz and welded acid ash-flow tuff clasts. The formation is unconformable on the Abitagua Granite (Jurassic), well exposed in the Río Zuñag [813250 9845000] (Fig. 2). The Napo Formation or Napo Group (Jaillard et al., 1997) (Albian–Maastrichtian) appears to overly the Hollín conformably. In the Río El Tigre [813500 9846050] it comprises several hundred metres of black calcareous bituminous mudstones, calc-siltstones and calcarenites with scattered coarse shelly limestones up to 20 m thick. The formation is overlain by the continental Tena Formation (Maastrichtian–Palaeocene?), more than 500 m of mottled red and green mudstones, siltstones, psuedo-oolitic, lithic and calcareous are-

nites and a basal unit of well sorted quartz arenite. The arenite contains about 5% blue quartz grains. The lithic arenites comprise mainly intraformational mudstone/siltstone clasts, with scattered shell fragments including oysters.

5.3. *Margajitas Formation (Tschopp, 1948)*

A stratigraphical problem, exacerbated by reverse faulting and overturning, exists between the Cosanga and sub-Andean faults (Figs. 2 and 3). The strata, which exceed 1 km in thickness, comprise dark grey, pyritous, and non-calcareous strongly cleaved silty mudstones and mudstones. In some places, the cleaved mudstones are light green or red. The dark grey mudstones include scattered units of massive, well-sorted quartz arenite up to 15 m thick. The arenites are either dark grey, because of interstitial organic material and pyrite, or, where oxidized, buff. Bioturbation, including horizontal traces on bases, is widespread. The arenites contain about 1% distinctive blue quartz grains and are strongly indurated; they contain many quartz tension gashes.

The Margajitas mudstones and quartzites, Palaeozoic in Tschopp (1948)'s opinion, were included in the Upano unit, a meta-andesite-dominated Jurassic sequence, by Litherland et al. (1994). However, the Margajitas mudstones include conformable, non-tectonised sequences, up to about 25 m thick, of cleaved limestone with typical Cretaceous bivalves and echinoid faunas (Figs. 2 and 3). The Margajitas quartz arenites are petrographically identical to Hollín arenites and contain a similar proportion of blue quartz. Therefore, we prefer to view the Margajitas Formation as a thick, basinal equivalent of the Hollín, Napo and, possibly, Tena formations of the Oriente Basin (cf. Bristow and Hoffstetter, 1977; personal communication, G. Ruiz, October 2002). The Cosanga Fault was clearly a major syndepositional fault and defined the E edge of the basin (Figs. 2 and 3). Furthermore, the sparse volcanic rocks in the Baños corridor probably belong to the Misahuallí Formation. The Río Upano (Fig. 1) also comprises phyllites and 'meta-greywackes with blue quartz grains' (Litherland, 1987), suggesting that exposures mapped as 'Upano unit' probably belong to the Margajitas Formation.

6. Intrusive igneous rocks (from W to E)

The Baños corridor includes a probable Late Triassic granitoid, but is dominated by Jurassic plutons, probably the roots of a major magmatic/volcanic arc. From W to E the plutons comprise:

6.1. Ulba plutonic complex

This is a ca. 2-km-wide belt of metamorphosed diorite, quartz diorite and granodiorite (Fig. 4). No isotopic dates are available. The rocks are remarkably homogeneous and show granoblastic (metamorphic) and igneous textures in thin section. The main lithology is a light grey, poorly foliated meta-quartz diorite with albite-oligoclase and minor quartz, biotite, epidote, chlorite, muscovite and garnet, locally altered to chlorite. Grain size is about 1–2 mm; scattered pink euhedral garnets attain 3 mm. Local quartzofeldspathic lenses define compositional banding. Northwest of Ulba Bridge [789600 9846150] (Fig. 4), there is a width of over 1 km of meta-quartz diorite with no obvious magmatic or tectonic fabric.

The E contact of the Ulba Plutonic Complex is the Ulba Shear Zone (Section 7.5). The W contact, probably intrusive, is well exposed in the Río Pastaza [788300 9846250]. It comprises interfingering meta-diorite and pelite (Fig. 10D). This is the mapped location of the Baños fault, which lies at the heart of a 2-km-wide shear zone and terrane boundary (Litherland et al., 1994). However, we find no difference in deformation style between the proposed 2-km-wide zone and the surrounding areas. The pelites are typical graphitic phyllites and schists with lenticular and folded quartz veins. The final 100 m before the contact comprises grey albite and garnet porphyroblastic schists with scattered black graphitic schists and rare psammites. The adjacent Ulba intrusions are massive or only weakly foliated and preserve igneous textures. We interpret the contact as intrusive, rather than a fault.

6.2. Tres Lagunas Granite (Late Triassic–Early Jurassic) (Litherland et al., 1994)

This crops out between Agoyán Bridge and Agoyán waterfall, a width of about 2 km (Figs. 2, 3 and 4). It is probably the most unmistakable lithology

of the Ecuadorian Andes and discontinuous outcrops are found from Colombia to Perú. Narrow slivers also occur within the Alao-Paute Unit, Inter-Andean Valley (Litherland et al., 1994), and Pujilí Fault (Hughes and Pilatasig, 2002) (Fig. 1). Radiometric data from southern Ecuador gave an errorochron of 200 ± 12 Ma by Rb–Sr and a more precise age of 227.6 ± 3.2 Ma by Pb isotope analysis of zircons (Litherland et al., 1994). The strongly deformed varieties of granite are described below (Section 7.2). Where least deformed, it is a light grey, coarse-grained porphyritic meta-granite comprising 20% euhedral to anhedral alkali feldspar phenocrysts, commonly smoky and concentrically zoned, up to 15 mm long, within a fine-grained granoblastic groundmass of quartz and albite-oligoclase. Litherland et al. (1994) record local rapakivi overgrowths on plagioclases, but we did not see these at Baños. Scattered anhedral blue quartz phenocrysts are up to 6 mm long. Muscovite (up to 5%) and biotite (up to 5%) are everywhere present, but amphibole is locally present in place of some of the biotite. Pinkish-red accessory garnets of unknown composition occur in scattered grains up to 0.5 mm. We have been able to identify very few examples of magmatic state fabrics. Only above Agoyán waterfall (Fig. 4) is sub-vertical magmatic banding defined locally by differences in grain size and mica content. The banding occurs at up to 30° to the tectonic foliation (seen in plan) and also shows igneous ‘cross-bedding’. Deformed pegmatite dykes (plagioclase+amphibole+spinel) and pelite xenoliths also occur here.

Litherland et al. (1994) mapped both contacts of the granite as major faults in the Baños corridor. The western is well exposed beneath Agoyán Bridge, a crucial locality for the geological history of the Cordillera Real (Fig. 7). The granite is fine-grained and strongly foliated, with common oval augen of smoky alkali feldspar up to 2 mm long. The augen have tails of recrystallized alkali feldspar and quartz. At the contact with pelites, the granite shows a finer-grained, quartz-rich, margin, 20 to 150 mm wide. The regional tectonic foliation cuts the contact obliquely and is weaker in this margin (Fig. 7). The adjacent pelites show a very weakly foliated halo, 50 to 150 mm wide, of >20% coarse-grained chlorite, probably after biotite. Farther from the contact, the pelites become moderately foliated and rich in albite and

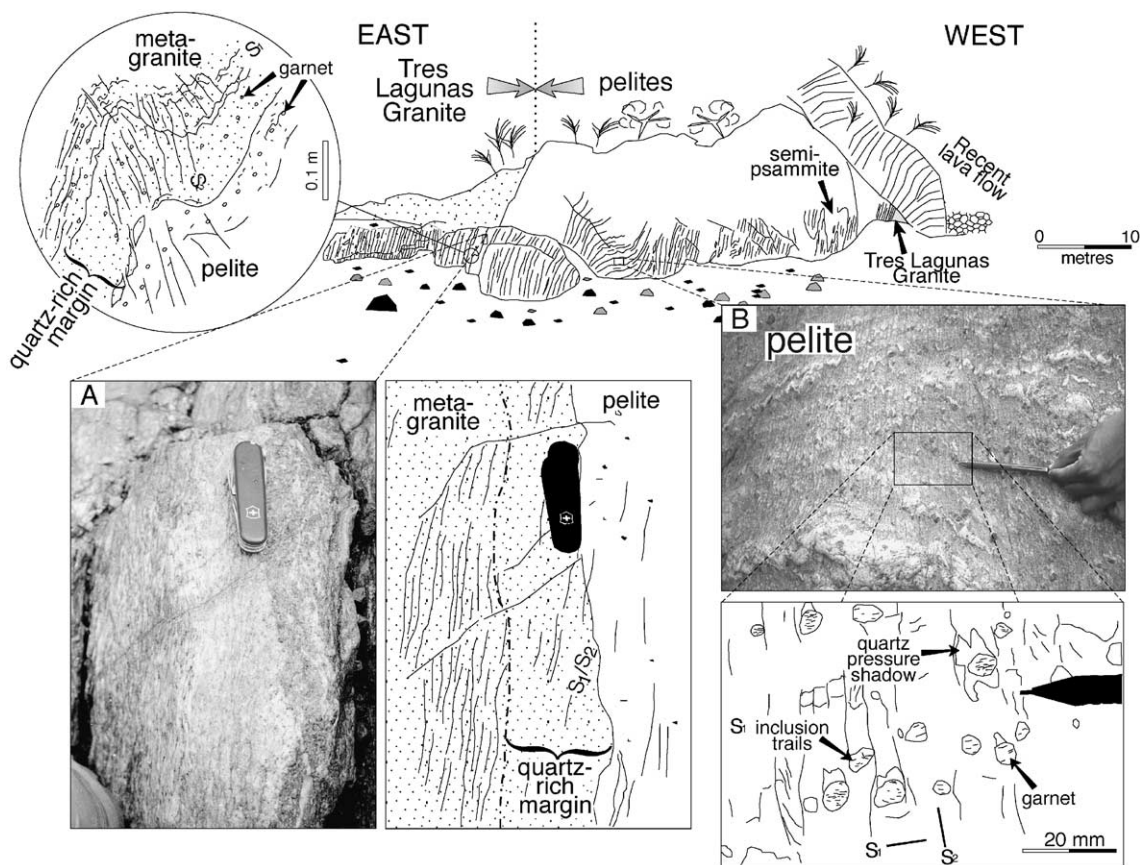


Fig. 7. Sketches and photographs of exposures at the contact between the Tres Lagunas Granite and pelites. Agoyán Bridge [791520 9845250].

garnet porphyroblasts. Across the contact, the percentage of garnet varies, from ca. 1% in the pelites, ca. 3% in the aureole, ca. 2% in the quartz-rich granite, to trace quantities in the main body of the granite. About 20 m W of the granite/pelite contact, there is another contact between pelite and Tres Lagunas Granite (Fig. 7), probably an apophysis of the main body. We interpret the western contact of the Tres Lagunas Granite as an intrusive contact, with minor metasomatism, and the quartz-rich band as a chilled margin. We see no evidence of a fault or metamorphosed unconformity.

The eastern contact of the Tres Lagunas Granite is reasonably well exposed in the Río Pastaza [793075 9845160]. Both the granite and adjacent pelites show a strong foliation for several hundred metres on either side. However, the critical 1 or 2 m around the contact are poorly exposed and dirty. A fault cannot

be ruled out, but an intrusive contact seems more likely. A second body of Tres Lagunas Granite, about 250 m wide, is exposed in the Baños-Puyo road [794300 9845440] (Fig. 5). It comprises foliated meta-granite with scattered blue quartz augen up to 10 mm long. The W contact is transition into greenschist and biotite schist of uncertain protolith. The E contact is the Río Blanco Fault (Fig. 5).

6.3. Azafrán Plutonic Complex (Late Jurassic?–Early Cretaceous) (= Azafrán unit of Litherland et al., 1994)

This multiple intrusion has an outcrop width of ca. 10 km (Figs. 3 and 5). The main lithologies are moderately foliated coarse-grained metamorphosed diorite, quartz diorite, and granodiorite. Original igneous textures, including granophyric texture, are

preserved widely and xenoliths are common. Magmatic state fabrics have not been recognized. Blue quartz phenocrysts occur at Machay [802405 9844930] (Fig. 5). The bulk of the eastern part of the outcrop is weakly foliated, to non-foliated, porphyritic granitoid with 3% biotite/amphibole and traces of epidote in a fine, or granophyric, groundmass. K–Ar and Rb–Sr isotopic data give a wide spread of ages for the pluton. Litherland et al. (1994) concluded that the most precise date for the plutonic complex is 142.7 ± 1 Ma, based on U–Pb concordia from zircons.

The W contact is with pyritous pelites and meta-sandstones. Litherland et al. (1994) mapped it as the ‘Llanganates fault’, a terrane boundary (Fig. 2). Exposures on the S bank of the Río Pastaza and in the Baños-Puyo road, both show narrow exposure gaps (<50 m). There is no evidence of unusual deformation or tectonic lenses of unusual lithologies. A fault cannot be ruled out, but we prefer to interpret it as a metamorphosed intrusive contact or unconformity.

6.4. Abitagua Granite (Middle Jurassic) (Sauer, 1950)

This is part of a chain, over 500 km long, of potassic Jurassic granites in the eastern Cordillera Real and sub-Andean Zone (Fig. 2). The chain includes the Zamora Granite in S Ecuador. The granite is pink and granophyric and shows no evidence of tectonic foliation or metamorphism. It is overlain unconformably by the Hollín Formation (Figs. 2 and 3). The granite has been dated by Rb–Sr at 173 ± 5 Ma (Baldock, 1982) and 162 ± 1 Ma (Aspden et al., 1991). A steep sliver of Abitagua Granite(?), with cataclastic deformation, occurs in the hanging wall of the Cosanga Fault and is overlain by the Margajitas Formation (Fig. 3).

6.5. Other intrusive igneous rocks

Folded and boudinaged amphibolite dykes are common in the Tres Lagunas Granite (Figs. 8 and 9) and within the Ulba Shear Zone (see below). They also occur within the Alao-Paute Unit at Las Juntas (Fig. 4) and upstream of Lligua [784260 9846050]. Meta-andesite/dolerite dykes are common in the Azafrán Plutonic Complex, e.g. Río Cristal

[803750 9843240]. In strongly foliated zones these dykes become epidotic greenschists, e.g. [802775 9844190].

7. Structure and metamorphism

The principal structures and foliations in the Baños corridor strike N–S and are commonly sub-vertical (Figs. 2, 3, 4 and 5). This is true of most of the Cordillera Real, though there are areas with flat-lying nappes and foliation farther N (Litherland et al., 1994).

7.1. Folds

The igneous contacts, bedding, compositional banding and foliations in the Río Pastaza are mainly N-striking, steeply dipping and homoclinal (Fig. 3). However, small-scale parasitic folds and rootless intrafolial folds are widespread. Textural evidence implies that they are F_2 , but see discussion below. Most are tight to isoclinal, with a wavelength of 1 m to 5 m. They are everywhere focused on competence contrasts, such as quartz veins, dykes or marbles within pelites (Figs. 4A,B and 10), or amphibolite/greenschist dykes within the plutons (Fig. 8). Most fold axes plunge gently or moderately S. There is a weak great circle distribution of fold axes on the stereogram, broadly corresponding with the N-striking S_2 foliation (Fig. 11). The vergence of minor folds suggests the existence of major fold closures and is shown in Figs. 3, 4 and 5. Litherland et al. (1994) believed that the amphibolite and pegmatite dykes in the Tres Lagunas Granite were not deformed, but there is abundant evidence, such as tight folds, boudinage and cross-cutting foliation (Figs. 8 and 9), that they were.

7.2. Foliations

Strain was strongly partitioned; this is reflected in the intensity and distribution of the tectonic foliations. There are two penetrative tectonic foliations (S_1 and S_2), and a localized non-penetrative crenulation (S_3), in the pelitic and meta-volcanic rocks of the Baños corridor (from Río Patate to the sub-Andean Fault)

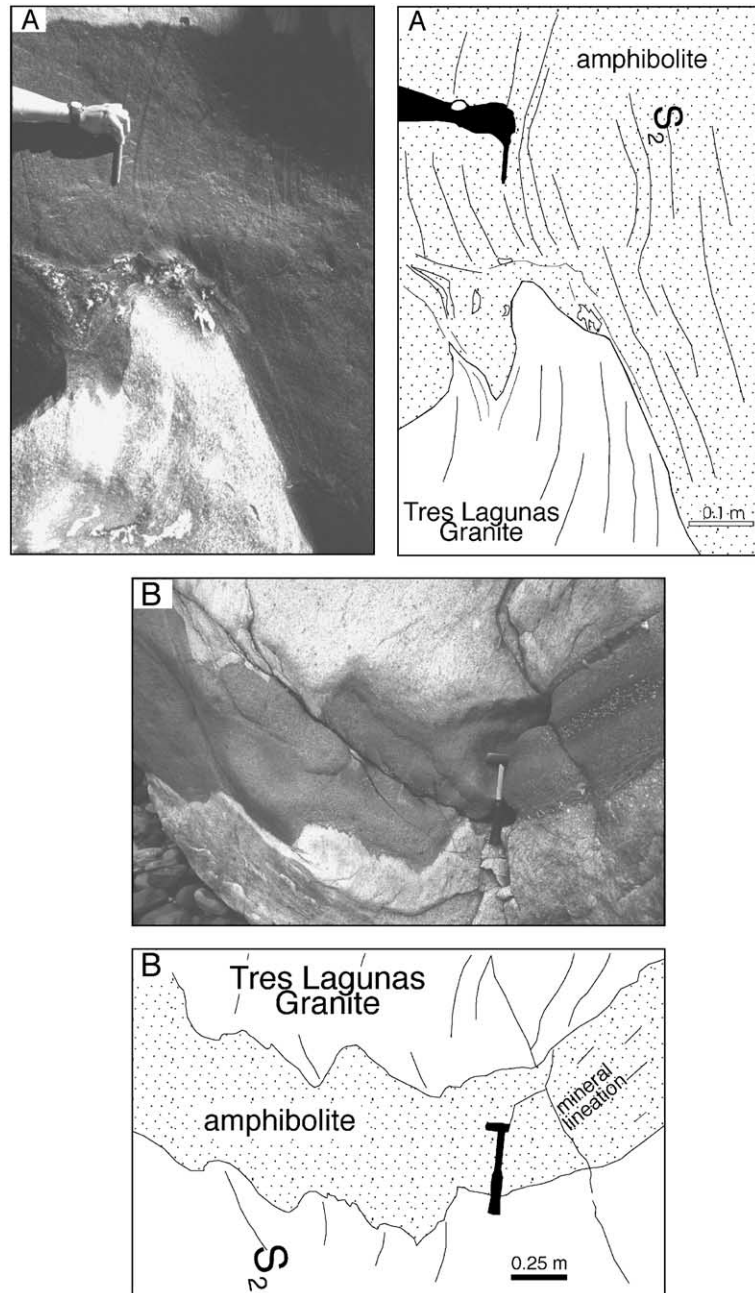


Fig. 8. (A) Folded contact between Tres Lagunas Granite and amphibolite dyke. (B) Folded amphibolite dyke with shallow mineral lineation. Above Agoyán waterfall [792720 9844950].

(Fig. 2). However, the competent units, such as the plutonic rocks and massive greenstones, responded very differently and commonly display a single foliation.

7.2.1. Metamorphosed sedimentary and volcanic rocks

Most pelites appear to show single strong foliation, comprising sub-millimetric mica-rich and mica-poor

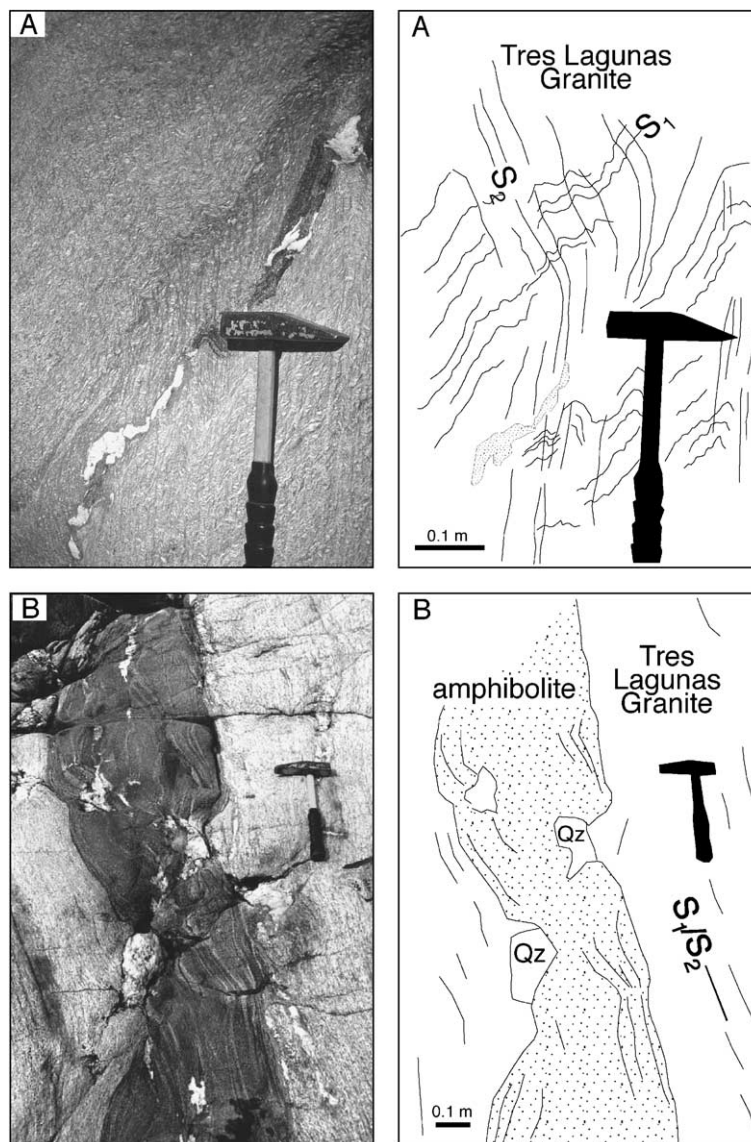
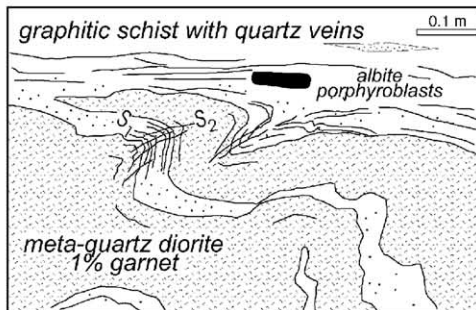
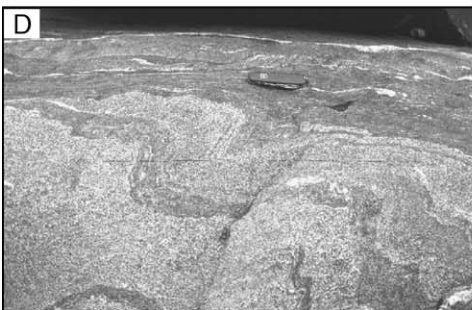
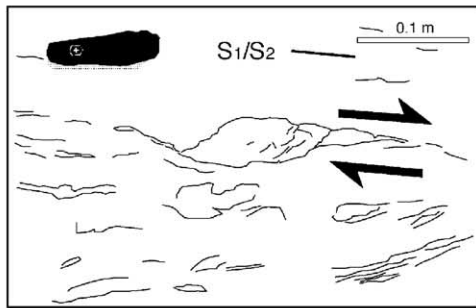
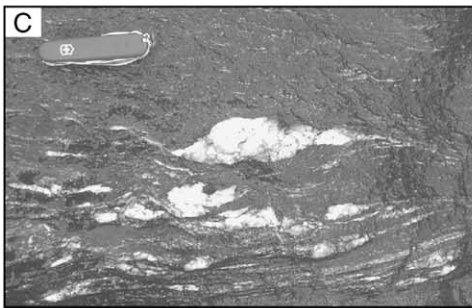
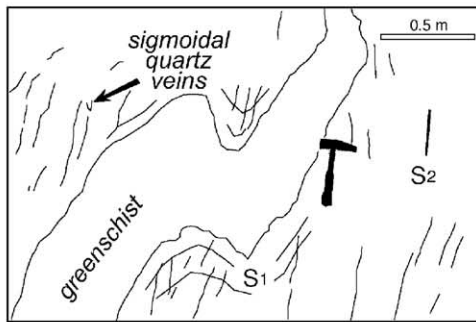
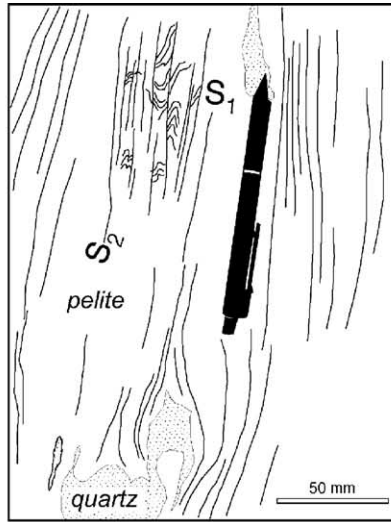


Fig. 9. (A) Tres Lagunas Granite with two overlapping tectonic foliations. Above Agoyán waterfall [792720 9844950]. (B) Boudinaged amphibolite dyke in Tres Lagunas Granite. Río Pastaza [793050 9845150].

bands. Where bedding can be identified, it is sub-parallel to this foliation. However, there is clear evidence of an early foliation preserved within parasitic fold hinges and porphyroblasts. This indicates that the principal foliation is S_2 . The S_1/S_2 relation-

ships are well preserved at Agoyán Bridge, in pelites adjacent to the Tres Lagunas Granite (Fig. 7). They also occur across the Cordillera Real, in lithologies such as greenschist, chloritic phyllites (former tuffites?), psammites, and graphitic phyllites (Figs.

Fig. 10. (A) Pelites with S_1/S_2 relationship. Río Pastaza [787980 9846450]. (B, C) Folded greenschist dyke and sigmoidal quartz veins within graphitic phyllites. Veined surface dips S at about 40° . Río Pastaza [793350 9845040]. (D) Intercalated meta-quartz diorite (Ulba Plutonic Complex) and graphitic schist. Río Pastaza [788120 9846715].



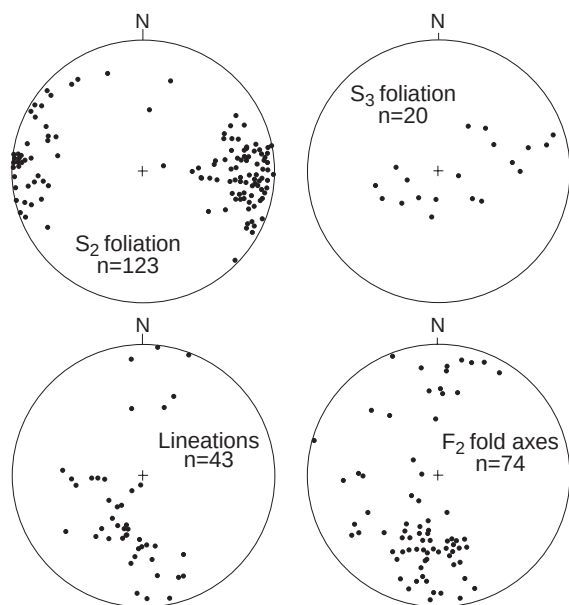


Fig. 11. Stereographic data from the Río Pastaza section (between Las Juntas and the sub-Andean Fault). Equal area projections. Minor fold axes are of folded bedding, compositional banding and quartz veins.

4A,B, 5 and 10). Lenticular quartz veins are common in the most ductile rocks, such as graphitic phyllite and schist. Boudinaged epidote-rich layers are common in the greenschists. The veins and boudins mostly developed during S_1 and it is important to emphasize that their shapes were subsequently modified, or folded, during formation of S_2 (Figs. 4A,B and 10A–C). Since they originate from polyphase, or a single progressive, deformation, the use of sigmoidal veins as regional kinematic indicators (cf. Litherland et al., 1994) is not valid.

In many places, especially at Agoyán Bridge (Fig. 7) and at the W contact of the Ulba Plutonic Complex, garnet and albite porphyroblasts preserve S_1 as delicate inclusions trails of graphite, mica and quartz (Fig. 12). This foliation developed at lower metamorphic grade; the constituent grains are finer and graphite is preserved. Incorporation of complete, non-rotated crenulations and microfolds indicates that the garnet and albite porphyroblasts passively overprinted crenulations of S_1 in the early stages of D_2 . With continued deformation, the porphyroblast shapes became modified and developed tails of granoblastic quartz. Outside the porphyroblasts, S_1 was rotated and

enhanced by coarse S_2 muscovite and quartz growth. Thin section sketches by Sheppard and Bushnell (1933), of greenschists from the S of the Cordillera Real, also depict porphyroblasts and augen with conserved S_1 fabric. This suggests that the S_1/S_2 relationship seen in the Baños corridor is widespread in the Cordillera Real.

The foliation in the Baños corridor therefore comprises S_1 , probably a low greenschist or sub-greenschist slaty cleavage, overprinted, transposed into, and largely destroyed by S_2 . There might be a component of an enhanced bedding parallel compaction fabric (S_0) in the pelites, but the occurrence of S_1 in igneous rocks (Figs. 5A and 10D) and plutons (see below) demonstrates that it is a true regional tectonic foliation.

The regional (S_1/S_2) foliation shows a decline in intensity towards the W, in the region of Patate and Las Juntas (Fig. 4). This gradual decline, and the lack of a clear-cut boundary between metamorphic and non-metamorphosed rocks, occurs along the entire W flank of the Cordillera Real; it has caused profound problems of stratigraphic correlation in the past, most noticeably N of Cuenca (Bristow, 1973). The situation is more clear-cut in the E, where the sub-Andean Fault marks a sharp drop in metamorphic grade and intensity of the foliation (Fig. 5). Sub-greenschist mudstones with a single, slaty cleavage occur between the sub-Andean and Cosanga faults (Figs. 2 and 3). The correlation of the cleavage with the core of the Cordillera Real is unclear, but it may well correlate with S_2 .

7.2.2. Plutonic rocks and massive lavas

We do not recognize magmatic state foliations (sensu Paterson et al., 1989) in the plutons. For example, there are no obvious phenocryst or crystal alignments in outcrop. However, we have not carried out an exhaustive petrographic study of the less deformed plutons, where such fabrics might be conserved. Dangerfield (in Litherland, 1989) examined a suite of granitoids from the Cordillera Real and believed that the foliations are tectonic and ‘cataclastic’. We agree that they are tectonic, but prefer dynamic recrystallization, which is consistent with the greenschist metamorphism. Corrosion of phenocrysts in a direction perpendicular to the foliation suggests that pressure solution was an important mechanism.

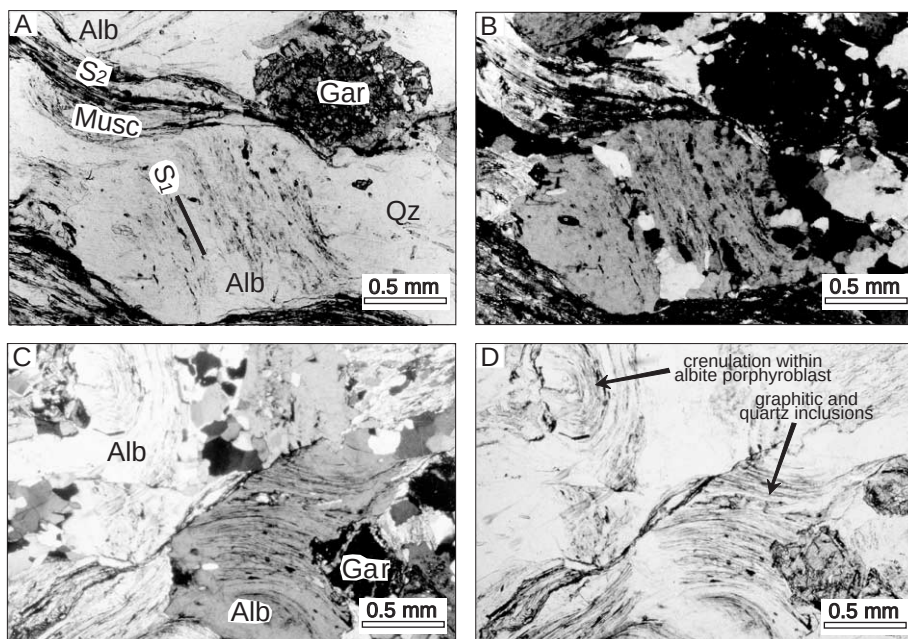


Fig. 12. Porphyroblasts of albite and garnet with S_1 inclusion trails. (A, B) Pelites 50 m from the intrusive contact with the Ulba Plutonic Complex. [788250 9846250]. (C, D) Pelites 20 m from the intrusive contact with the Tres Lagunas Granite. [791520 9845250].

Some parts of the Ulba, Tres Lagunas and Azafrán plutons are only weakly foliated and preserve primary igneous textures. Where least deformed, for example in the Río Pastaza [792056 9845033], the Tres Lagunas Granite is a meta-granite and shows a recrystallized granoblastic fine-grained groundmass. Some alkali feldspars are euhedral and retain concentric compositional zonation, but most have suffered some recrystallization and grain size reduction at the margins, rendering them ragged. Poorly defined domains of biotite, muscovite, and amphibole mark the foliation. Amphibolite dykes are folded and the foliation passes directly from the granite into a weaker foliation within the dykes (Fig. 8). Other parts of plutons show strong mylonitic foliations with intense grain size reduction and alternating quartz-ofeldspathic and micaceous domains, each less than 0.5 mm wide. Anastomosing and lenticular granoblastic quartz and muscovite domains separate augen of relict igneous texture and phenocrysts. Litherland et al. (1994) described S-C fabrics (cf. Lister and Snoke, 1984) and believed that they reflect strong simple shear, of strike-slip origin.

Amphibolite dykes in the strongly foliated plutons are elongated and boudinaged parallel to the foliation (Fig. 9B); rare xenoliths have aspect ratios of up to 10:1, with long axes parallel to the foliation (the extensional axis cannot be determined). This clearly indicates strong shear of some sort. However, with the exception of a few narrow shear zones (<1 m) that are frequently bound on one side by a fault, we have seen little evidence of strong simple shear. Apparent S-C fabrics occur in only a few places, most notably over about 75 m of the immediate hanging wall of the sub-Andean Fault. Elsewhere, there is frequently no discernible asymmetry to the tectonic fabric and it is best described as ‘anastomosing’. There is also no apparent relationship between orientation and strength of foliation, which might support development in a zone of simple shear. The areas of strong foliation commonly have ill-defined margins, with only a gradual increase in foliation intensity from both directions. Furthermore, overlapping, and unrelated, tectonic foliations have probably been interpreted in the past as S-C mylonites, for example, the Tres Lagunas Granite above Agoyán waterfall (Fig. 9A). We therefore suggest that many of the strong

foliations reflect pure shear (flattening), though in the absence of strain markers we cannot be sure.

The correlation of foliations between the plutons and pelites is rarely clear. However, the best evidence comes from the contact regions. Beneath Agoyán Bridge, a strong foliation is defined within the Tres Lagunas Granite by millimetric compositional bands of sugary quartz and feldspar and scattered alkali feldspar augen to 3 mm, separated by very narrow mica domains (0.1–0.2 mm wide). This foliation crosses the contact into pelites (Fig. 7). However, the granite/pelite contact and the foliation are folded and cut by a penetrative foliation that shows strong new mica growth. Furthermore, careful examination of the granite shows a second foliation, defined by muscovite mica growth, at an angle of about 5° to the fine compositional banding. We therefore interpret the first foliation (compositional banding) as S_1 and the second as S_2 . Another line of evidence comes from the margins of deformed dykes within the plutons (Figs. 5A and 10D). Invariably, they show a S_1 tectonic foliation, crenulated by S_2 , close to the contact. Such fabrics have been noted both within the dykes and the plutons in these contact areas and

support a common deformational history for the pelites and plutons.

7.3. Lineations

A lineation occurs in many of the foliated rocks. Its understanding is crucial to the tectonic interpretation of the Cordillera Real. Litherland et al. (1994) suggest that it indicates an important component of strike-slip.

In the Baños corridor the lineation mainly plunges about $30\text{--}50^\circ\text{S}$ in both weakly and strongly foliated rocks and is everywhere parallel to minor fold axes (Fig. 11). In the metamorphosed sedimentary rocks it is formed by the intersection of S_1 and S_2 and by crenulations/microfolds of S_1 and compositional bands (Figs. 10A and 13). The stereograms show that fold axes and lineations are distributed along the same great circles, equivalent to the mean S_2 foliation (Fig. 11). The lineations in the plutons are much less well developed and their origins more varied. Where two foliations can be identified, such as Agoyán Bridge (Fig. 7), the lineation is an intersection lineation. Elsewhere, it is clearly a mineral stretching lineation, defined by elongate quartz, feldspar and mica. The

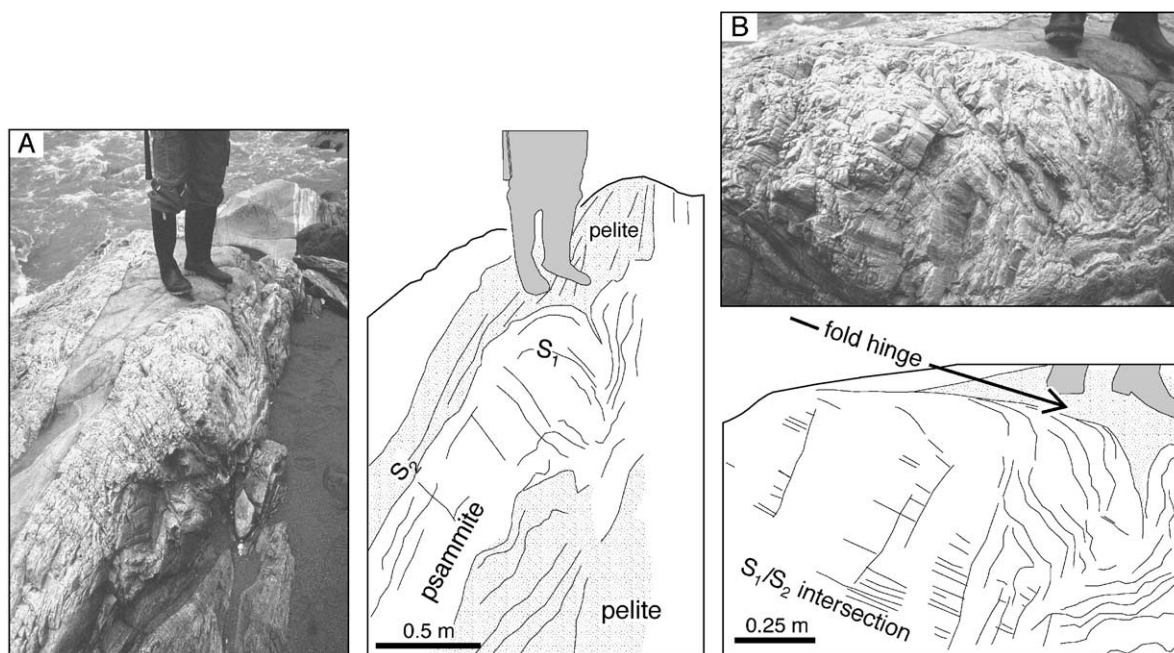


Fig. 13. (A, B) Folded psammite beds with strong S_1/S_2 intersection lineation parallel to the fold axis. Río Pastaza [786980 9846450].

best mineral lineations occur in the E part of the Azafrán Plutonic Complex and pitch steeply southwards within the foliation (Fig. 5). These rocks have a strong component of sub-vertical elongation and contain common augen with quartzofeldspathic tails, e.g. Machay [802450 9844950]. A weak mineral lineation, defined by amphiboles plunging S at about 36° , is apparent in a folded amphibolite dyke in the Tres Lagunas Granite above Agoyán waterfall (Fig. 8B). It plunges broadly parallel to the fold axis. Pressure shadows locally form lineations; those on garnets at Agoyán Bridge, within pelites beside the Tres Lagunas Granite contact, consistently pitch sub-vertically in the S_1/S_2 foliation (Fig. 7B).

7.4. Kinematic indicators

Potential kinematic indicators include parasitic folds, S-C textures and deformed pebbles. Fold axes in the Baños corridor can generally be measured and vergence recorded (Figs. 3, 4, 5 and 11). Z and S geometries dominate and are shown in Fig. 3; they imply the existence of major S-plunging fold closures. Undoubted S-C fabrics are rare and in some cases apparent S-C fabrics probably represent overlapping foliations belonging to different deformations. A weak S-C fabric within Tres Lagunas Granite immediately above Agoyán waterfall (Figs. 4 and 8) implies reverse movement. Folded and sigmoidal quartz veins (Figs. 4A,B and 10A–C), confined to pelites, and with a mainly shallow to moderate S plunge (Fig. 11), have also been used as kinematic indicators. Litherland et al. (1994) presented photographs of S-C fabrics and sigmoidal quartz veins to support dextral strike-slip movements in the Cordillera. S-C fabrics, with a mix of dextral and sinistral displacements, are also described from foliated Tres Lagunas Granite and Jurassic quartz-sericite schists at the Peggy gold deposit in southern Ecuador (PRODEMINCA, 2000, Vol. 4). However, we do not believe these are reliable, because of polyphase deformation. Furthermore, sigmoidal veins commonly change in asymmetry across a single exposure, or from limb to limb of a fold (Fig. 4A). They largely reflect local tectonic conditions, such as simple shear around F_2 folds due to flexural slip.

Fabric analysis provides the most reliable indicator of the principal axes of the finite strain ellipsoid.

Centre-to-centre fabric analysis (Ramsay, 1967, pp. 195–197) of deformed pebbles at the confluence of the ríos Blanco and Pastaza (Fig. 5) indicates a plane-strain type strain ellipsoid, with the extensional axis inclined about 70° to the N.

7.5. Faults and shear zones

Significant faults exist in the Baños corridor. Some clearly have apparent vertical displacements of many kilometres, probably with a major young (Miocene–Pliocene) component, and have juxtaposed outcrops with very different metamorphic grade. From W to E they comprise:

7.5.1. La Playa Fault

This cuts the Alao-Paute Unit and calcareous pelites. Its trace is marked by abundant landslips (Figs. 2 and 4). The fault, exposed in the Río Pastaza [783600 9845150], comprises several fault strands of rock gouge and smeared out black calcareous phyllite. Greenschists crop out between the strands. Slickensides pitch steeply and indicate a dip-slip component of displacement (Fig. 4). Non-deformed Tertiary (?) diorite dykes up to 8 m wide were intruded into the fault zone and show chilled margins.

7.5.2. Ulba Shear Zone

This ductile shear zone, at least 30 m wide, but no more than 100 m, marks the E contact of the Ulba Plutonic Complex (Fig. 4). The sheared rocks comprise rusty weathered, garnetiferous paragneisses/schists with strongly boudinaged and folded amphibolites. The best exposures occur at Ulba Bridge (Fig. 4); others are probably concealed by a Recent lava. S_1 and S_2 foliations are recognized within the zone and fold axes are strongly curvilinear. Kinematic indicators are contradictory and no coherent displacement direction emerges.

7.5.3. Río Blanco Fault

A significant fault with a major vertical component is inferred at Río Blanco (Figs. 2, 3 and 5) to explain an abrupt decline in metamorphic grade, from high greenschist in the W, with garnet and albite porphyroblasts, to low greenschist, chloritoid-dominated, in the E.

7.5.4. Sub-Andean Fault

This major E-verging fault marks the E limit of greenschist facies metamorphic rocks (Figs. 2, 3 and 5). It is clearly visible, along with the Cosanga Fault, on the recent digital terrain models of Ecuador and the Oriente (Baby and Souris, 2001; Souris, 2001). The hanging wall comprises foliated Azafrán granitoid; the footwall comprises sub-greenschist cleaved mudstones (Margajitas Formation). A reverse component of several kilometres is implied. The fault zone comprises several strands and contains wedges of sheared diorite [805270 9844200], probably a portion of Azafrán Plutonic Complex, and welded ash-flow tuff (Misahuallí Formation). Unequivocal S-C textures and shear zones occur in the hanging wall, e.g. [805150 9843900]. The S-C intersection, which marks the intermediate axis of the finite strain ellipsoid, plunges at 45° towards 020°. The S-C fabrics indicate reverse, E-verging oblique slip movement (Fig. 5).

7.5.5. Cosanga fault

This E-verging fault was an important syndepositional fault (Section 5.3). It probably has a Tertiary (Miocene–Pliocene?) reverse dip-slip component of several kilometres. It brings cleaved rocks over non-metamorphosed, locally bituminous, Cretaceous–Paleocene rocks in the footwall (Figs. 2 and 3). A small lens of sheared (Abitagua?) granite occurs in the hanging wall [810400 9844125]. Cretaceous stratigraphic links suggest that it is not a terrane boundary (cf. Litherland et al., 1994).

7.5.6. Abitagua fault

This major E-verging fault is active, thrusting the Jurassic Abitagua Granite over Quaternary gravel terraces (DGGM, 1980; personal communication Severine Bes de Berc, July 2002), and is marked by a major fault scarp.

7.6. Metamorphism and deformation

There appear to be two phases of regional metamorphism in the main Cordillera Real, though a progressive metamorphism and deformation is a strong possibility. Initially, low- or sub-greenschist conditions accompanied S_1 slaty cleavage development. Later, mid- to high greenschist metamorphism

began with porphyroblast (albite-oligoclase, garnet, chloritoid, biotite) growth during crenulation of S_1 . This developed into S_2 formation, defined by growth of much coarser grained quartz, muscovite and biotite. Unpublished microprobe data from 19 Baños samples by Tomas Feininger (personal communication, December 2001) indicate that garnets within the pelites are dominantly almandine, typically 50–75%, with 20–30% grossularite. Pressure shadows of quartz are developed around the garnets (Fig. 7B). Garnet appears about 2 km W of Baños and continues about 7 km E. Chloritoid dominates in some pelites, implying a strongly aluminous protolith; at Río Blanco (Fig. 5), it forms up to 60% by volume. The main metamorphic minerals in the Alao-Paute Unit are chlorite, actinolite, albite, quartz, biotite and epidote; fine-grained biotite is developed as far W as Las Juntas (Fig. 4). Higher-grade metamorphism, possibly up to low amphibolite in the Ulba Shear Zone because of gneissic textures, occurred within the core of the Cordillera Real. Latterly, the regional metamorphism appears to have had a strong thermal (static) component. It caused widespread granoblastic recrystallization of quartz and growth of randomly oriented coarse muscovite in the pelites, resulting in very tough rocks that are difficult to split. Garnet is commonly altered to chlorite, suggesting, along with epidote and chlorite in the greenschists, widespread retrogression during the final stages of regional metamorphism. Substantial late (Tertiary to Recent) movements on major faults, such as the sub-Andean and Río Blanco faults, subsequently juxtaposed rocks of different metamorphic grade (Fig. 3).

8. Interpretation

8.1. Stratigraphy

A stratigraphic scheme for the Cordillera Real and sub-Andean Zone is still a long way off. However, potential correlations can be identified. They are aided by the recognition that the Cordillera has no obvious terrane complications and that it comprises an eroded and uplifted Palaeozoic core, flanked by younger (Jurassic–Cretaceous) sequences.

Firstly, the Misahuallí Formation (Jurassic) of the Oriente almost certainly equates with the volcanic

rocks of the sub-Andean Zone ('Upano unit' of Litherland et al., 1994). Geochemical work by Romeuf et al. (1995) suggests that the Misahuallí is the remnant of a Jurassic calc-alkaline arc, generated by eastward subduction beneath the S American continent. The Jurassic plutons of the sub-Andean Zone and Cordillera Real are probably the roots of the arc. Secondly, the Margajitas Formation is clearly Cretaceous and equivalent to the relatively condensed sequence of the Oriente Basin. It therefore marks a major sedimentary depocentre in the sub-Andean Zone, probably controlled by down-to-the-W subsidence on the Cosanga Fault. This basin received influxes of well-sorted, platform-type quartz sand. The overall configuration of the sub-Andean Zone is very similar to the profound changes in facies and thickness from the Llanos Basin to the Eastern Cordillera of Colombia (Cooper et al., 1995) and from the Marañon Basin to northern Andes of Perú (Cobbing, 1972; Jaillard et al., 2000). The W margin of the Margajitas basin is not obvious; the reverse movement on the sub-Andean Fault has uplifted and eroded potential equivalents. However, pyritous pelites and meta-sandstones with blue quartz grains at Río Blanco (Section 4.1), in the heart of the Cordillera Real, may be an outlier of the basin.

Stratigraphic links between the Inter-Andean Valley and Oriente are suggested by detrital blue quartz. To our knowledge, these distinctive grains are not known from clastic rocks older than the Jurassic/Cretaceous in Ecuador. From W to E, blue quartz occurs in: quartzites and greywackes in the Inter-Andean Valley; greywackes in the Río Patate; meta-sandstones at Río Blanco; Margajitas Formation quartz arenites; Hollín and Tena quartz arenites. It also occurs in meta-quartzite beds within major graphitic phyllite sequences S of Saraguro and E of Quilanga, in S Ecuador (Zamora Series of Baldock, 1982) (Fig. 1). Two fossiliferous localities occur in phyllites close to Saraguro; one has Jurassic–Cretaceous pollen, the other Middle Jurassic–Early Cretaceous spores (Litherland et al., 1994).

The source of the blue quartz is uncertain. However, the Hollín Formation of the Oriente thins westward (Jaillard et al., 1997) and is probably sourced from the Guyana Shield. Paleocurrent observations, including our own, support this (Jaillard et al., 2000). However, Ruiz et al. (2002) suggested

multiple Hollín sources on the basis of varied detrital zircon fission track age populations. The increase in blue quartz in the well-sorted basal Tena Formation arenite, perhaps westerly or axially derived, suggests that Hollín or Margajitas sandstones were eroded and recycled. Reworked coarse shells in the lower Tena might represent eroded Napo material. Blue quartz-bearing Mesozoic plutons in the northern Andes, such as the Tres Lagunas Granite, Azafrán Plutonic Complex, and Portachuela pluton in northern Perú (our observations), are also potential sources.

8.2. Tectonism and timing of deformation

There is little tectonic evidence in the Baños corridor of terranes; similar deformation histories occur across the Cordillera. The constant S_1/S_2 relationship in the pelites is an important unifying link. There was, however, strong partitioning of strain; the pelites everywhere suffered strong shortening, whereas the plutons, because of their rigidity, suffered inhomogeneous deformation, local simple shear (Ulba Shear Zone), or no discernible deformation at all.

The polyphase history makes it difficult to determine the overall style of deformation in the Baños corridor. Conventional methods, such as cleavage transection of folds (Sanderson and Marchini, 1984; Pratt and Fitches, 1993), do not work because of strong flattening. The suggestion that gently plunging lineations indicate strike-slip or transpression (Litherland et al., 1994) is untenable because they are largely intersection lineations. The gently plunging minor folds and steep extensional axes, indicated by deformed pebbles and porphyroblast tails instead suggest an orthogonal, upright deformation, though transpression cannot be ruled out. We interpret the minor folds as parasitic to a major periclinal anticline, which forms the (Palaeozoic) core of the Cordillera Real (= 'Loja terrane' of Litherland et al., 1994). Southwards, the core closes near Río Paute (Fig. 1), and the Paleozoic pelites and Tres Lagunas Granite pinch out (Litherland et al., 1994).

Triassic magmatism and local metamorphism are recorded in the northern Andes (reviewed in Jaillard et al., 2000). Triassic rifting is also known from the Oriente Basin (Dashwood and Abbotts, 1990). However, there is little evidence in the Baños corridor that the Tres Lagunas Granite was intruded into an active

shear zone or that it suffered strike-slip or transpressional deformation (Aspden and Litherland, 1992; Aspden et al., 1992; Litherland et al., 1994). The Tres Lagunas Granite lacks a Triassic, syn- to late-emplacement, foliation. Furthermore, the amphibolite dykes and pegmatites within it are not syn-tectonic (cf. Litherland et al., 1994), but pre-tectonic. The foliation in the plutonic rocks has no obvious magmatic or high-temperature subsolidus characteristics; instead it formed by dynamic recrystallization and grain size reduction. Increased intensity of foliation at pelite/pluton contacts simply indicates strong contact strains, rather than major terrane boundaries.

Litherland et al. (1994) suggested that the Tres Lagunas Event caused local crustal melting; they believed that the Sabanilla migmatites of SE Ecuador are related to the Tres Lagunas Event. However, no transition from Tres Lagunas Granite to migmatite is recorded in the Cordillera Real. Indeed, the Sabanilla migmatites pass across the Peruvian border into Precambrian gneisses (INGEMMET, 1995). Rafts of Precambrian gneisses also occur a short distance to the E, in the Zamora batholith (Litherland et al., 1994). Furthermore, Palaeozoic and Precambrian gneisses and amphibolites occur in the Cordillera Real at Papallacta (Herbert, 1977; Litherland et al., 1994) (Fig. 1). These occurrences of high-grade rocks may be inliers of an extensive Precambrian basement, analogous to that in SE Colombia (cf. the Central Cordillera, Garzón Massif and Sierra de la Macarena; e.g. Kroonenberg, 1982).

We agree with Litherland et al. (1994) that the principal deformation of the Cordillera Real, their 'Peltetec Event', occurred either in the Early or Late Cretaceous, prior to deposition of the Tena Formation red-beds (Maastrichtian). The latter were probably derived from the rising Cordillera Real (Baldock, 1982). Litherland et al. (1994) preferred an Early Cretaceous age for the deformation, whereas we prefer a Late Cretaceous age. The principal age constraints are as follows: (1) within the Baños corridor, penetrative foliation affects the following: pre-Late Triassic (Palaeozoic) pelites; Late Triassic–Early Jurassic plutons (Tres Lagunas Granite); Jurassic plutons (Azafrán Plutonic Complex); Cretaceous–Paleocene sedimentary rocks (Margajitas Formation). (2) Elsewhere in the Cordillera Real, foliated rocks are

cut by non-deformed Palaeocene plutons (Litherland et al., 1994). (3) The K–Ar systems of plutons were reset at the Cretaceous/Tertiary boundary (Feininger, 1975, 1982; Litherland et al., 1994). (4) A major unconformity occurs at the base of the Hollín Formation (Aptian–Albian) in the Oriente Basin and in the sub-Andean Zone (Dashwood and Abbotts, 1990). (5) Early Cretaceous sandstones overlie deformed, but non-metamorphosed, Alao-Paute rocks in the Cebadas Valley, on the E flank of the Inter-Andean Valley (personal communication, Étienne Jaillard, April 2003).

The Early Cretaceous unconformity clearly represents an important tectonic event. However, we suggest it represents not orogeny, metamorphism, and penetrative deformation, but block faulting of the Jurassic magmatic and volcanic arc, followed by peneplanation. This process is evident in the Oriente, where Early Cretaceous block faulting was accommodated by N–S basement faults (Dashwood and Abbotts, 1990). In the sub-Andean Zone, the Jurassic Abitagua Granite was unroofed, presumably from several km depth, and Misahuallí volcanic rocks were locally tilted into a steep orientation. However, there was no penetrative deformation of the Abitagua Granite, the Misahuallí or Chapiza formations. Furthermore, significant compressive deformation in the Oriente Basin, probably an excellent litmus test for events in adjacent areas, began only in the Campanian–Maastrichtian (Dashwood and Abbotts, 1990). Likewise, significant metamorphic detritus began to arrive in forearc turbidites of the Cordillera Occidental only in the Paleocene–Eocene (e.g. Toro Álava and Jaillard, 2002). Older clastic sequences, such as the Yunguilla Formation (Campanian–Maastrichtian), are dominated by volcanic arc material. There is also evidence of a Campanian or Maastrichtian switch from turbiditic sedimentation to shallow water conditions in the Yunguilla Formation at Cumbe, on the W flank of the Cordillera Real (Pratt et al., 1997) (Fig. 1).

We therefore believe that the main deformation of the Cordillera Real occurred at the Napo-Tena formation transition (Maastrichtian). The Cordillera emerged and sedimentation in the Oriente switched to red-beds (Baldock, 1982). The Coniacian–Maastrichtian interval also saw the onset of compressional deformation and uplift of the Peruvian Andean margin

and marine regression in the Andean Basin (Steinmann, 1929; Jaillard et al., 2000). Compressive deformation continued into the Tertiary, in pulses (Spikings et al., 2000, 2001), accommodated along the sub-Andean faults. These movements raised the core of the Cordillera orogen, effectively removing the W margin of the Cretaceous Margajitas Basin and making it impossible to calculate the original western extent.

9. Conclusions

We present an informal stratigraphic and structural framework for the Cordillera Real as a basis for future work. Great uncertainties remain, particularly the contacts between Palaeozoic and Jurassic/Cretaceous pelites. We cannot rule out the existence of buried Palaeozoic or Precambrian terranes, as suggested in Colombia by Irving (1975). However, the similarities across the Cordillera Real far outweigh the differences. We prefer former interpretations that model the Cordillera as coherent autochthonous crust (Baldock, 1982; DGGM, 1982).

Inliers of high-grade metamorphic rock in the Cordillera Real may represent fragments of the Precambrian Guyana Shield basement, but none is exposed at the erosion level of the Baños corridor. The core instead comprises probable Palaeozoic pelites. These may correlate with Palaeozoic outcrops in the sub-Andean Zone of Ecuador, northern Perú, and Eastern Cordillera of Colombia (e.g. Quetame Massif; Grösser and Prössl, 1991). The Palaeozoic pelites were intruded by a Late Triassic/Early Jurassic granite (Tres Lagunas), reflecting Tethyan rifting (Aspden and Litherland, 1992). Major subduction-related magmatism followed in the Jurassic, broadly synchronous with the eruption of a subaerial to submarine calc-alkaline arc (Misahuallí Formation). Until radiometric dating and reliable geochemistry are available for the Alao-Paute Unit, we prefer to interpret it as a Jurassic submarine basalt/andesite sequence. A correlation with the Misahuallí is possible and should be investigated. Varied geochemistry, including island arc, is characteristic of marginal basins (Kokelaar et al., 1984). The possibility that the Cordillera Real comprises a marginal basin sited on thinned continental crust should be explored.

Because of subsequent uplift and deformation, it is hard to know if the Cordillera Real was a major depocentre during the Jurassic arc episode. However, deposition was probably strongly controlled by N–S basement faults and linear buoyant plutons (Abitagua and Azafrán). Strong extensional faulting (?) and peneplanation occurred in the Early Cretaceous (Hollín unconformity) in the Oriente, sub-Andean Zone, and possibly in the Inter-Andean Valley.

Blue quartz provides a possible link between the two sides of the Cordillera. It is present in Cretaceous rocks; it may also be present in Jurassic rocks. Fingerprinting of the quartz is required and provenance studies may well strengthen the links between the Inter-Andean Valley and the Oriente Basin.

The newly recognized Cretaceous basin in the sub-Andean Zone (and Cordillera Real?) provides a potential source area for the petroleum of the Oriente Basin. This is backed up by oil analyses from the Oriente; they indicate a Napo source, but the necessary volume of mature Napo strata is not found in the Oriente Basin. Like Feininger (1975) and Dashwood and Abbotts (1990), we suggest derivation from the now-metamorphosed rocks of the sub-Andean Zone (and Cordillera Real?). The Margajitas and Hollín arenites and Cosanga Fault probably provided for up-dip hydrocarbon migration. Without knowing the heat flow, it is difficult to judge how much burial would have been required to generate hydrocarbons; 3 or 4 km might be a realistic estimate. We suggest that this was achieved by an enormous thickness of Napo and possibly Tena-equivalent strata (Late Cretaceous), subsequently largely lost by erosion. Emplacement of early E-verging nappes is another possibility, but, in contrast to northern Perú, structures in the Ecuadorian sub-Andean Zone are generally not thin-skinned.

The boundary between oceanic and S American continental crust in Ecuador must lie W of the Cordillera Real. Our interpretation would place the suture at the Pujilí Fault (cf. Hughes and Pilatasig, 2002).

Collision between the oldest oceanic terrane (Pallatanga) (Kerr et al., 2002) and continental South America occurred in the Late Cretaceous and probably drove the deformation of the Cordillera Real. It is hard to envisage orogenic deformation in a non-collisional setting, which probably rules out subtle

changes in plate convergence rate and vector as causes. The profound shortening and nappe tectonics of the Cordillera Real imply a strong orthogonal component to this collision rather than strike-slip, though strike-slip may have been partitioned into the main suture (Pujilí Fault?). Other faults, such as the sub-Andean, probably suffered oblique/reverse displacements. This raises the issue of how the accreted oceanic terrane escaped deformation and metamorphism. The answer is probably twofold. Firstly, the Andean (marginal?) basins were sandwiched between two buttresses: the thick accreted oceanic plateau of the coastal plain (Guillier et al., 2001), and the Archaean crust of the Guyana Shield. Secondly, the basins were predisposed to deformation since they contained many kilometres of mud-dominated rocks and were sited on probable thinned continental crust.

The existence of large-scale E-verging nappes in the N half of the Cordillera (Litherland et al., 1994) suggests that the Baños corridor represents the deeper levels of the orogen and that the steep foliations and axial planes turn upwards and outwards, akin to a flower structure (cf. Harding, 1985). If this is the case, then there may have been a component of transpression in the deformation of the orogen.

The Late Cretaceous–Paleocene orogenic event marked the beginning of the Andean cycle of compressive deformation and fault-dominated tectonics. The flysch-like Tertiary deposits in the Oriente Basin accumulated during continued phases of compressional deformation (e.g. Baldock, 1982; Christophoul et al., 2002) and reverse faulting continues at present in the sub-Andean Basin.

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